

[BUG] [Return-types] Update the output shape of `qml.probs` for `default.q

<https://github.com/PennyLaneAI/pennylane/issues/1607>

ROW 100

[BUG] [Return-types] Update the output shape of `qml.probs` for `default.qubit.torch` #1607

New issue

Closed



antalszava opened on Aug 30, 2021

Expected behavior

Using `default.qubit.torch` and backpropagation, `return qml.probs(0), qml.probs(1)` returns a `(2, 2)` shape tensor (just as with TensorFlow).

Actual behavior

A `(4,)` tensor is returned for `return qml.probs(0), qml.probs(1)` in Torch+backprop. This could be due to differing behaviour in the `DefaultQubitTorch._asarray()` method.

Additional information

No response

Source code

```
import pennylane as qml

dev = qml.device('default.qubit.torch', wires=3)

@qml.qnode(dev, interface='torch')
def circuit():
    return qml.probs(0), qml.probs(1)

circuit()
```

```
tensor([1., 0., 1., 0.], dtype=torch.float64)
```

```
import pennylane as qml

dev = qml.device('default.qubit.tf', wires=3)

@qml.qnode(dev, interface='tf')
def circuit():
    return qml.probs(0), qml.probs(1)

circuit()
```

```
<tf.Tensor: shape=(2, 2), dtype=float64, numpy=
array([[1., 0.],
       [1., 0.]])>
```

Tracebacks

No response

System information

```
```shell
Name: PennyLane
Version: 0.18.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author: None
Author-email: None
License: Apache License 2.0
Location: /xanadu/pennylane
Requires: numpy, scipy, networkx, autograd, toml, appdirs, semantic-version, autoray, cachetools
Required-by: PennyLane-Cirq, PennyLane-Orchestra, PennyLane-Qchem, PennyLane-SF, pennylane-qulacs, PennyLane-
Platform info: Linux-5.11.0-27-generic-x86_64-with-glibc2.10
Python version: 3.8.5
```

## Assignees

antalszava

rmoyard

## Labels

bug

## Type

No type

## Projects

No projects

## Milestone

No milestone

## Relationships

None yet

## Development

No branches or pull requests

## Notifications

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## Participants



```
- cirq.qsimh (PennyLane-Cirq-0.17.1)
- cirq.simulator (PennyLane-Cirq-0.17.1)
- orchestra.forest (PennyLane-Orchestra-0.15.0)
- orchestra.ibmq (PennyLane-Orchestra-0.15.0)
- orchestra.qiskit (PennyLane-Orchestra-0.15.0)
- orchestra.qulacs (PennyLane-Orchestra-0.15.0)
- strawberryfields.fock (PennyLane-SF-0.16.0.dev0)
- strawberryfields.gaussian (PennyLane-SF-0.16.0.dev0)
- strawberryfields.gbs (PennyLane-SF-0.16.0.dev0)
- strawberryfields.remote (PennyLane-SF-0.16.0.dev0)
- strawberryfields.tf (PennyLane-SF-0.16.0.dev0)
- qulacs.simulator (PennyLane-qulacs-0.17.0.dev0)
- ionq.qpu (PennyLane-IonQ-0.17.0.dev0)
- ionq.simulator (PennyLane-IonQ-0.17.0.dev0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.4.1.dev0)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.4.1.dev0)
- forest.numpy_wavefunction (PennyLane-Forest-0.17.0.dev0)
- forest.qvm (PennyLane-Forest-0.17.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.17.0.dev0)
- honeywell.hqs (PennyLane-Honeywell-0.16.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.18.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.18.0.dev0)
- lightning.qubit (PennyLane-Lightning-0.18.0.dev0)
- default.gaussian (PennyLane-0.18.0.dev0)
- default.mixed (PennyLane-0.18.0.dev0)
- default.qubit (PennyLane-0.18.0.dev0)
- default.qubit.autograd (PennyLane-0.18.0.dev0)
- default.qubit.jax (PennyLane-0.18.0.dev0)
- default.qubit.tf (PennyLane-0.18.0.dev0)
- default.qubit.torch (PennyLane-0.18.0.dev0)
- default.tensor (PennyLane-0.18.0.dev0)
- default.tensor.tf (PennyLane-0.18.0.dev0)
```

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



antalszava added **bug** on Aug 30, 2021

github-actions assigned antalszava on Aug 30, 2021

antalszava mentioned this on Aug 30, 2021

[Update tests to ensure that Torch backpropagation is supported #1598](#)

josh146 mentioned this on Sep 21, 2021

[Add autodiff integration tests for the new QNode pipeline #1646](#)

antalszava mentioned this on Jul 29, 2022

[Use parameter broadcasting in param\\_shift #2749](#)



dwierichs on Aug 2, 2022

Contributor ...

An instance of where this bug became apparent/relevant is the use of broadcasting in `qml.gradients.param_shift`: Due to the differing output shape of the torch device in its `_asarray` method, the if-condition in `gradients/parameter_shift._evaluate_gradient` needs to check for torch devices and handles the corresponding case differently than for all other interfaces. This leads to self-consistent behaviour within the torch interface/device scenario, but showcases that the `_asarray` method should be adapted to yield consistent results with other devices or produce predictable output shapes instead of iteratively flattening and concatenating the returned values.



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Jaybsoni assigned rmoyard on Aug 2, 2022

AlbertMitjans assigned AlbertMitjans and unassigned AlbertMitjans on Aug 10, 2022

 Jaysoni assigned [rmoyard](#) on Aug 2, 2022

 AlbertMitjans assigned [AlbertMitjans](#) and unassigned [AlbertMitjans](#) on Aug 10, 2022



antalszava on Nov 29, 2022

Contributor Author ...

Note that this bug will no longer be relevant with the new return types:

```
In [1]: 1 import pennylane as qml
 2
 3 qml.enable_return()
 4
 5 dev = qml.device('default.qubit.torch', wires=3)
 6
 7 @qml.qnode(dev, interface='torch')
 8 def circuit():
 9 return qml.probs(0), qml.probs(1)
 10
 11 circuit()

Out[1]: (tensor([1., 0.], dtype=torch.float64), tensor([1., 0.], dtype=torch.float64))

In [2]: 1 import pennylane as qml
 2
 3 qml.enable_return()
 4
 5 dev = qml.device('default.qubit.tf', wires=3)
 6
 7 @qml.qnode(dev, interface='tf')
 8 def circuit():
 9 return qml.probs(0), qml.probs(1)
 10
 11 circuit()

Out[2]: (<tf.Tensor: shape=(2,), dtype=float64, numpy=array([1., 0.])>,
 <tf.Tensor: shape=(2,), dtype=float64, numpy=array([1., 0.])>)
```



 antalszava changed the title ~~[BUG] Update the output shape of `qml.probs` for `default.qubit.torch`~~ [BUG] [Return types] Update the output shape of `qml.probs` for `default.qubit.torch` on Nov 29, 2022

 antalszava changed the title ~~[BUG] [Return types] Update the output shape of `qml.probs` for `default.qubit.torch`~~ [BUG] [Return-types] Update the output shape of `qml.probs` for `default.qubit.torch` on Nov 29, 2022



albi3ro on Apr 4, 2023

Contributor ...

Given this bug is fixed by the new return types specification, I'm marking this bug as done.



 albi3ro closed this as [completed](#) on Apr 4, 2023

# Use parameter broadcasting in `param_shift` #2749

<> Code

Merged dwierichs merged 184 commits into master from parameter-broadcasting-4 on Aug 3, 2022

Conversation 82 Commits 184 Checks 0 Files changed 5

+1,400 -191



dwierichs commented on Jun 20, 2022

Contributor ...

This PR follows up on [#2627](#) and makes use of the newly introduced parameter broadcasting in `qml.gradients.param_shift`.  
Due to inconsistent output handling, it currently does not support tapes/QNodes with multiple return values and does not support shot vectors yet.



Reviewers

antalszava ✓  
mixd ✓

Assignees

No one assigned

Labels